CENTENNIAL COLLEGE

ETEC 326 (SEC.001): Technical Project

SKI SAFETY GPS

PROJECT DEMONSTRATION (GROUP 9)

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INTRODUCTION

The Problem:

- High Cost of Current Ski Safety Tech
- Limited by Cellular Coverage (unreliable on slopes)
- Subscription Fees

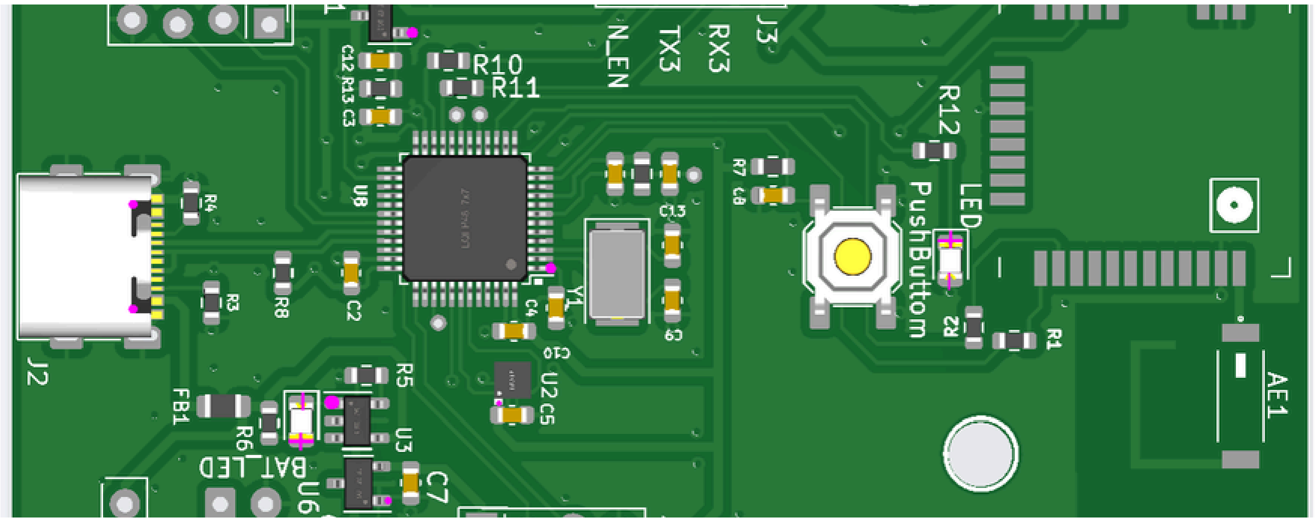
Our Solution:

- Compact GPS + LoRaWAN device
- Real-time Tracking
- Crash Detection
- Under $100 per Device
- No Subscription Fees

Target Users:

- Skiers
- Ski Resorts (need scalable safety solutions)

PREVIOUS INNOVATIONS



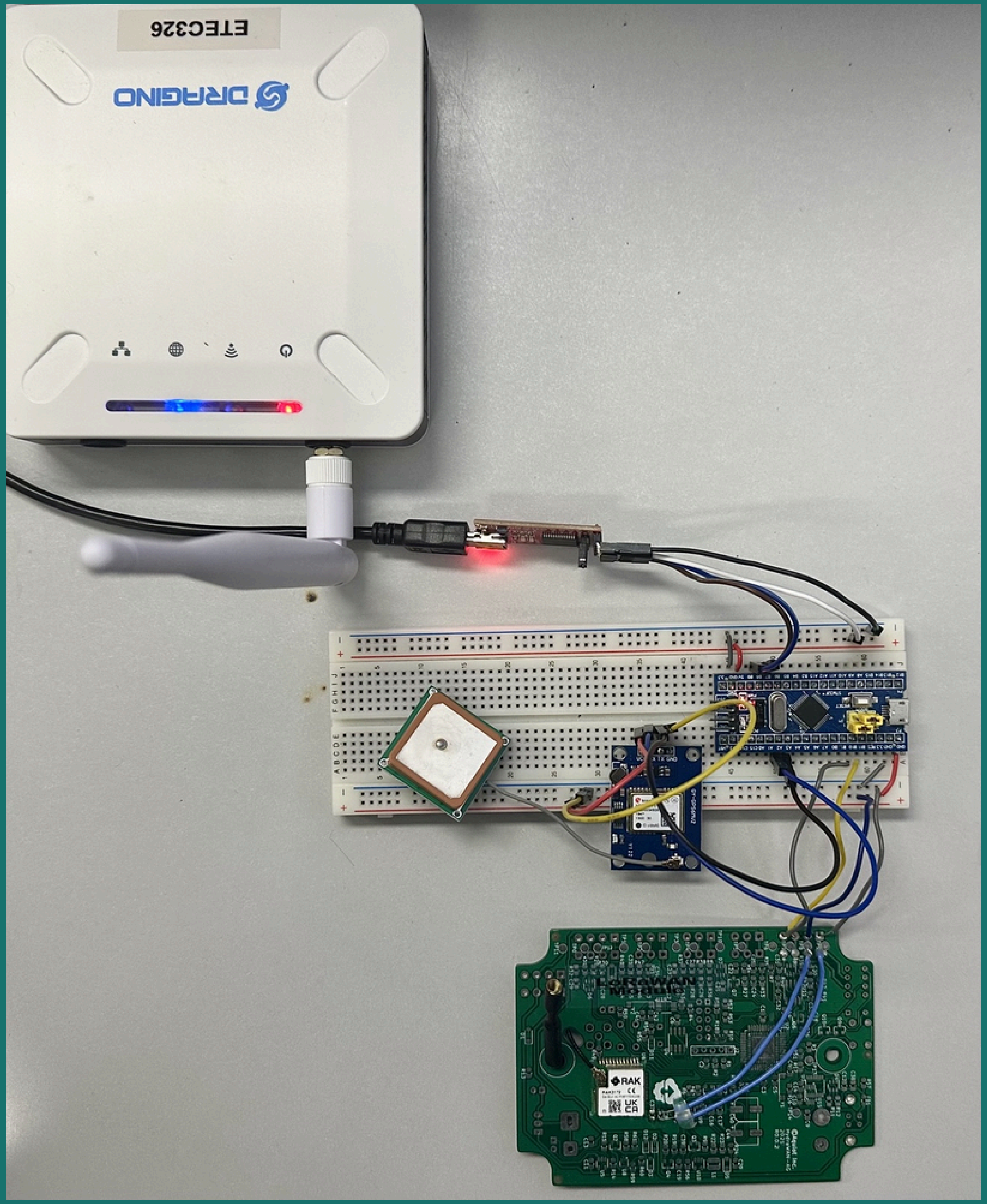
Elderly Safety

- “Individuals with dementia or Alzheimer's disease are susceptible to wandering or becoming disoriented, with a tendency to wander into areas where they may not be readily found. To address this issue, researchers propose an assistive LoRa-based tracking device that can be worn or attached to a wheelchair or walker for real-time location monitoring.”

Valuable Hardware

- LoRaWAN Module
- Accelerometer
- Custom PCB

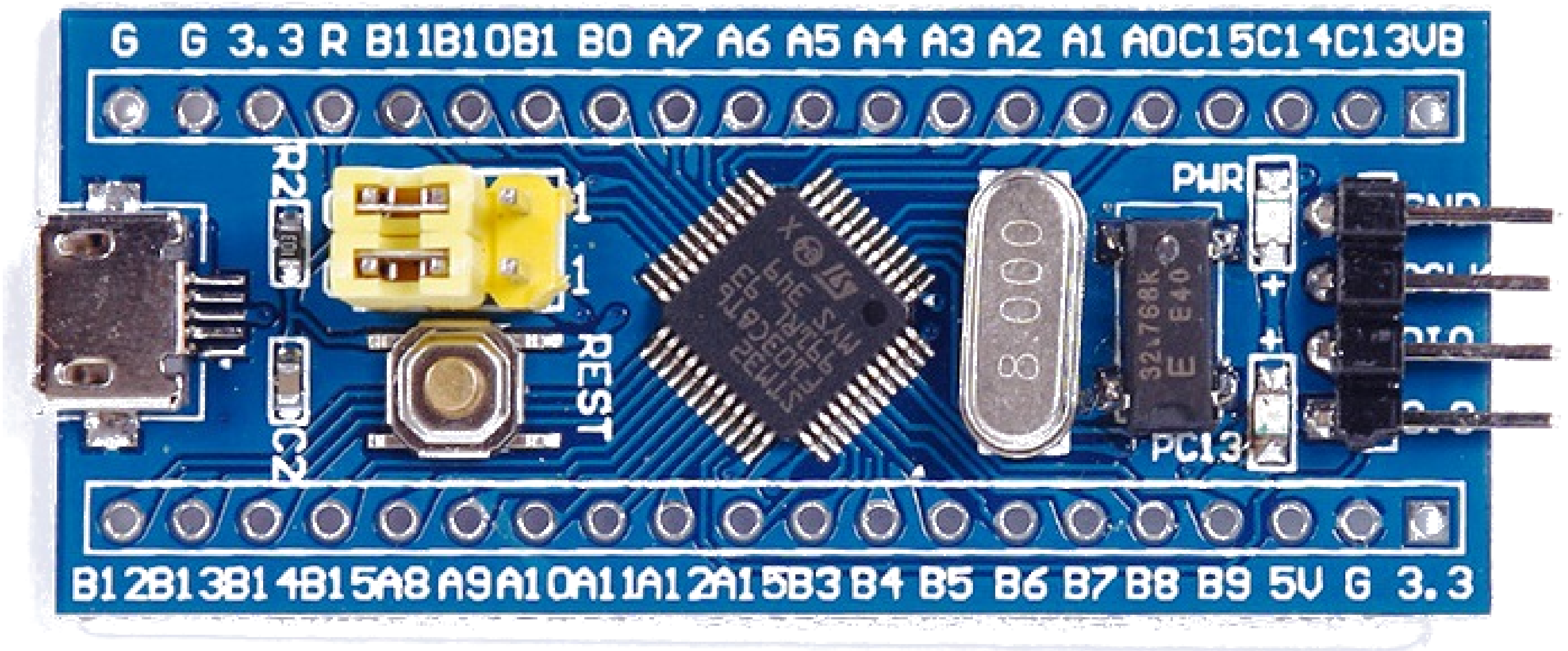
DEMONSTRATION



🔧 Hardware Components:

- STM32F103C6 Microcontroller
- RAK 3172
- Antenna
- LPS8N LoRaWAN Gateway
- NEO-8M GPS + LIS2DH12 Accelerometer

STM32F103C6 MICROCONTROLLER



Overview

- MCU Model: STM32F103C6 (ARM Cortex-M3)
- Core Speed: 72 MHz
- Flash Memory: 32 KB
- RAM: 10 KB
- Operating Voltage: 2.0V – 3.6V
- Package: LQFP48, QFN36

Key Features

✓ High-Performance Cortex-M3 Core

- Efficient for LoRaWAN stack processing

✓ Rich Peripherals:

- USART/SPI/I2C (For LoRa module communication, e.g., SX1276)
- 12-bit ADC (Sensor data acquisition)
- Low-Power Modes (Essential for battery-operated nodes)

✓ Development Support:

- STM32CubeMX (Pin configuration & code generation)
- Arduino-Compatible (Via STM32duino)

RAK3172

Overview

- Model: RAK3172 (LoRaWAN Module)
- Compatibility: Designed for LoRaWAN IoT devices
- **Range: Up to 15 km**
- LoRaWAN Class Support: Class A and C
- Power Supply: 1.8V – 3.6V (typically 3.3V), Ultra-low power (down to 1.69 µA in sleep mode)

Key Features

✓ LoRaWAN 1.0.3 Protocol Support

- Fully compliant with The Things Network and other LoRaWAN servers

✓ Ultra-Low Power Consumption

- Optimized for battery-powered applications (deep sleep mode supported)

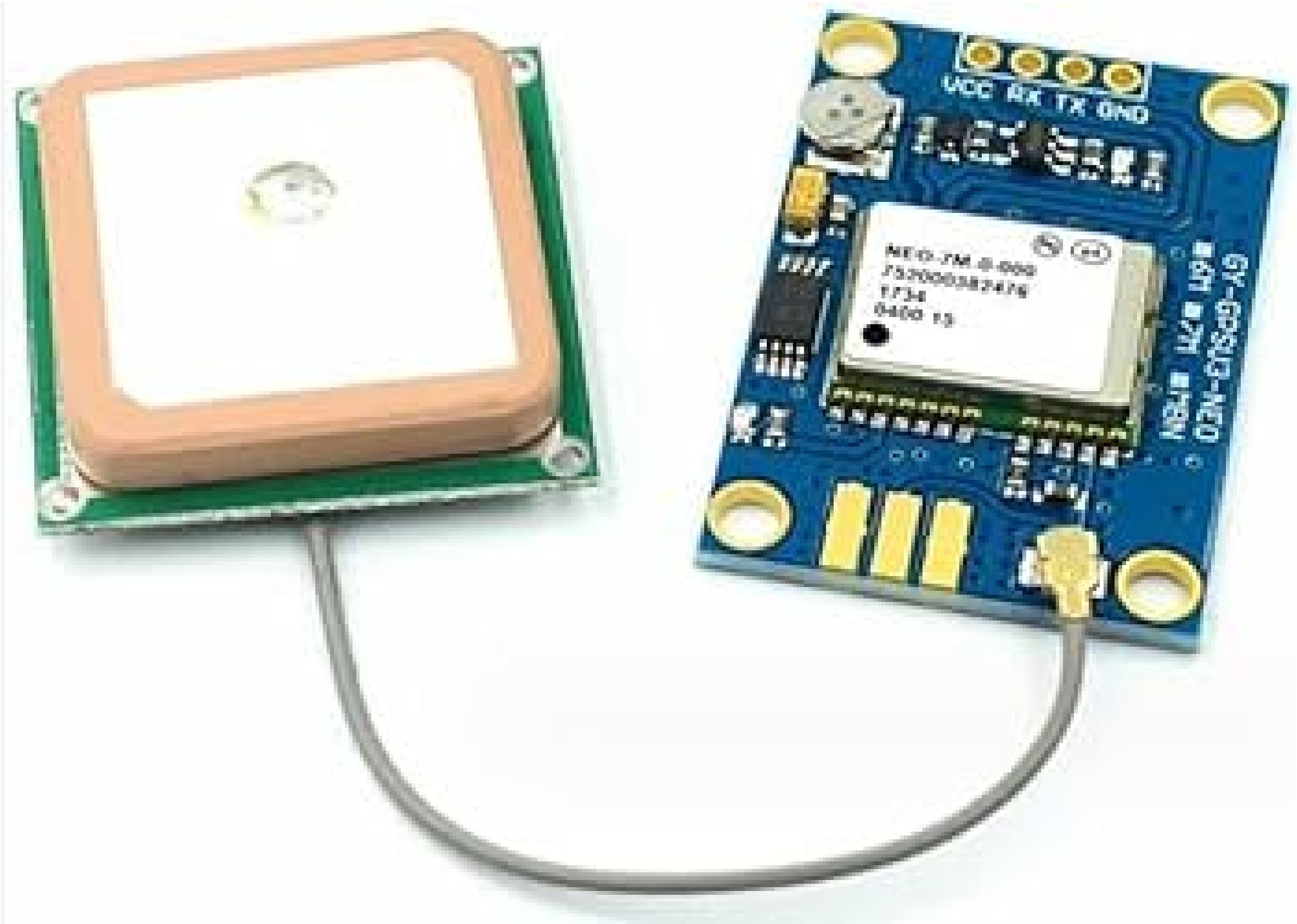
✓ Long-Range Communication

- Reliable transmission over several kilometers in open environments

✓ AT Command Interface

- Simplifies integration with microcontrollers via UART

NEO-8M GPS



Overview

- Model: u-blox NEO-8M (GNSS/GPS Receiver)
- Compatibility: Works with LoRaWAN trackers & IoT devices
- Positioning Accuracy: ~2.5m (GPS), <2m (with SBAS)
- Update Rate: Up to 10Hz (configurable)
- Power Supply: 3.3V DC, Low power consumption (~40mA active)

Key Features

✓ Multi-GNSS Support
- GPS, GLONASS, Galileo, BeiDou

✓ High Sensitivity (-167 dBm)
- Works in weak signal areas

✓ TTFF (Time to First Fix)
- <1 sec (Hot start), ~25 sec (Cold start)

✓ UART Interface
- Easy integration with MCUs (Arduino, STM32)

✓ Compact & Low-Cost
- Ideal for asset tracking & IoT applications

LIS2DH12

Overview

- Model: LIS2DH12
- Compatibility: Suitable fopr motion detection, and impact sensing
- Measurement Range: ±2g, ±4g, ±8g, ±16g (selectable)
- Update Rate: 1 Hz to 5.3 kHz (configurable)
- Power Supply: 1.71V – 3.6V, Low power modes supported (down to 2 µA in low-power mode)

Key Features

✓ 3-Axis Acceleration Sensing

- Captures movement along X, Y, and Z axes for accurate motion analysis

✓ Embedded FIFO & Interrupts

- Built-in motion detection, click detection, and free-fall recognition

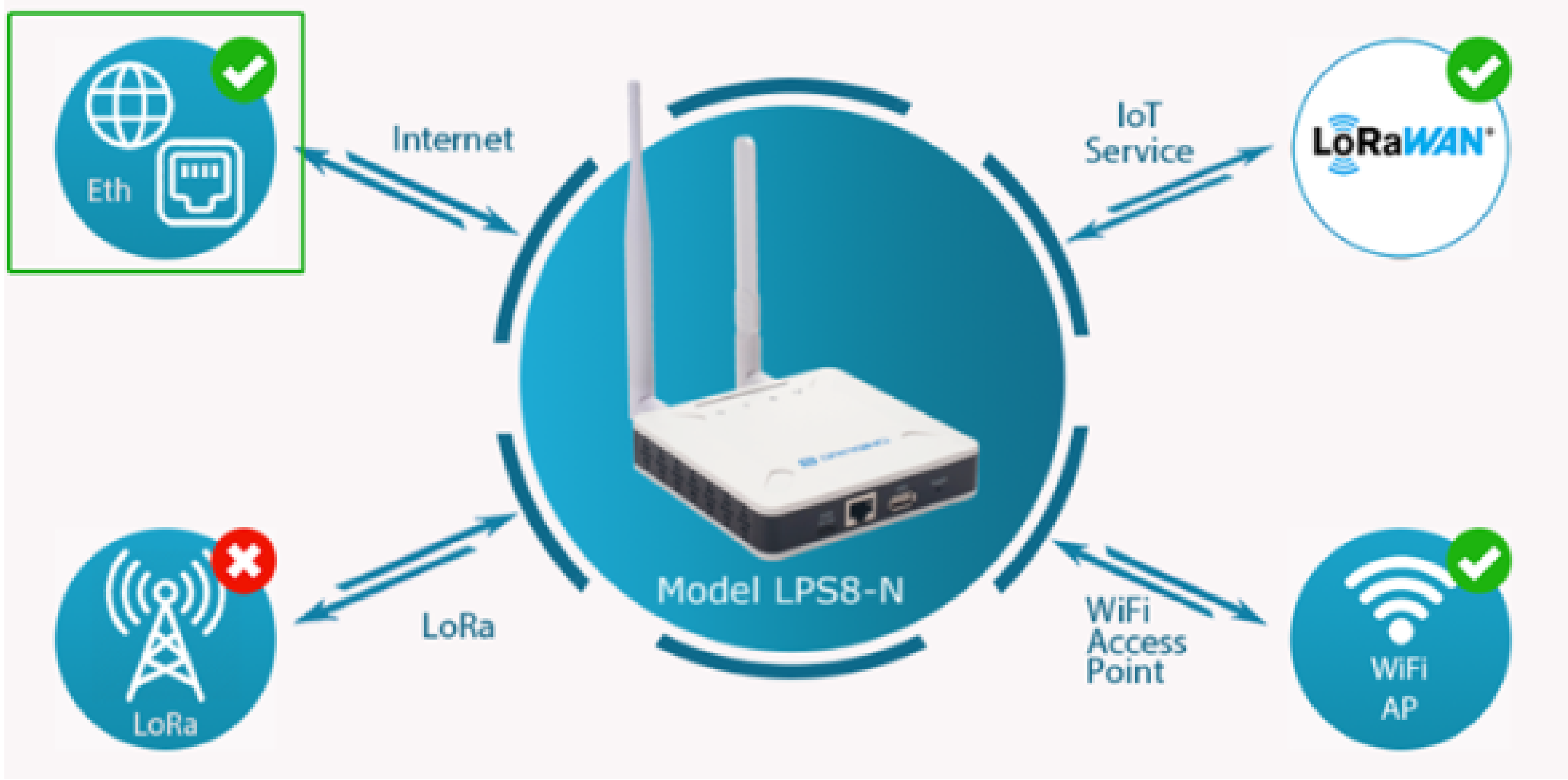
✓ Ultra-Low Power Operation

- Ideal for battery-operated and always-on devices

✓ High Sensitivity (Low Noise)

- Reliable performance in dynamic environments

LPS8N LORAWAN GATEWAY



Overview

Model: LPS8N (LoRaWAN Gateway)

Frequency: EU868 / US915 / AS923 (Region-specific)

Chipset: Semtech SX1308

Backhaul: Ethernet, Wi-Fi, Cellular (optional)

Power: PoE (Power over Ethernet) or 12V DC

Key Features

✓ 8-Channel LoRa Concentrator
- Supports multiple simultaneous transmissions

✓ High Sensitivity (-142 dBm)
- Long-range communication

✓ Indoor/Outdoor Use
- Rugged enclosure (IP67 optional)

✓ OpenWRT OS
- Flexible configuration & management

✓ Supports LoRaWAN Protocols
- Compatible with Class A/B/C devices

ADDITIONALLY FUNCTIONS

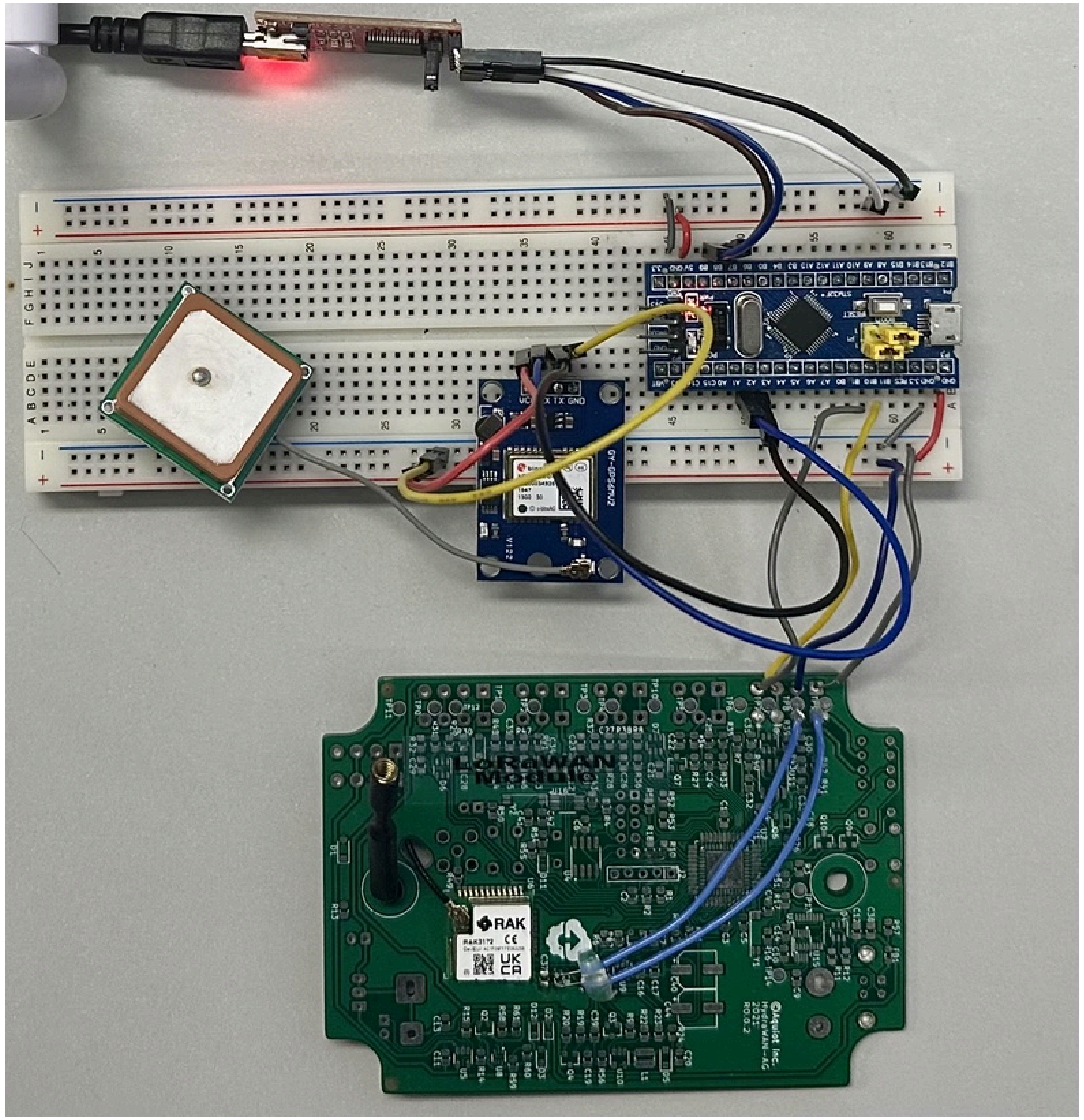
Pushbutton

- Manual report from the user that there is an issue

Power/Battery

- 3.3v DC required for PCB

WIRING



FTDI:

RX - PB6

TX- PB7

NEO8M:

RX-PA2

TX- PA3

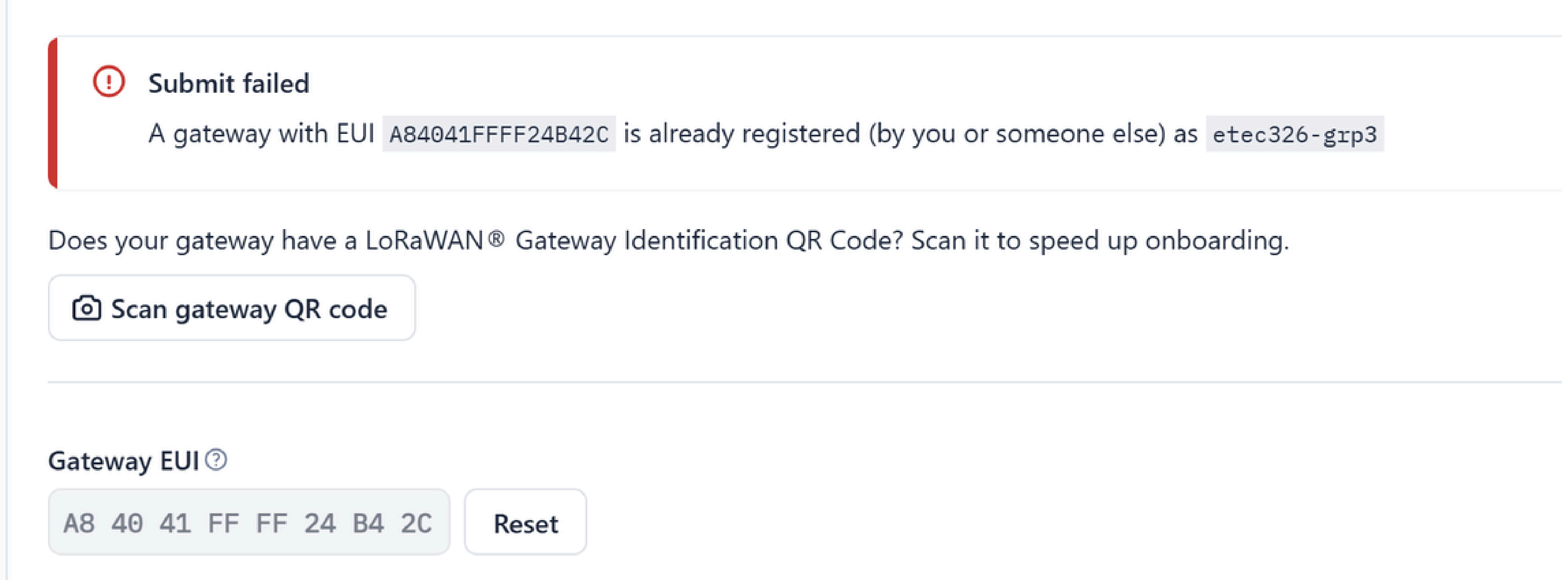
RAK3172:

RX -PB10

TX - PB11

TROUBLESHOOTING

01



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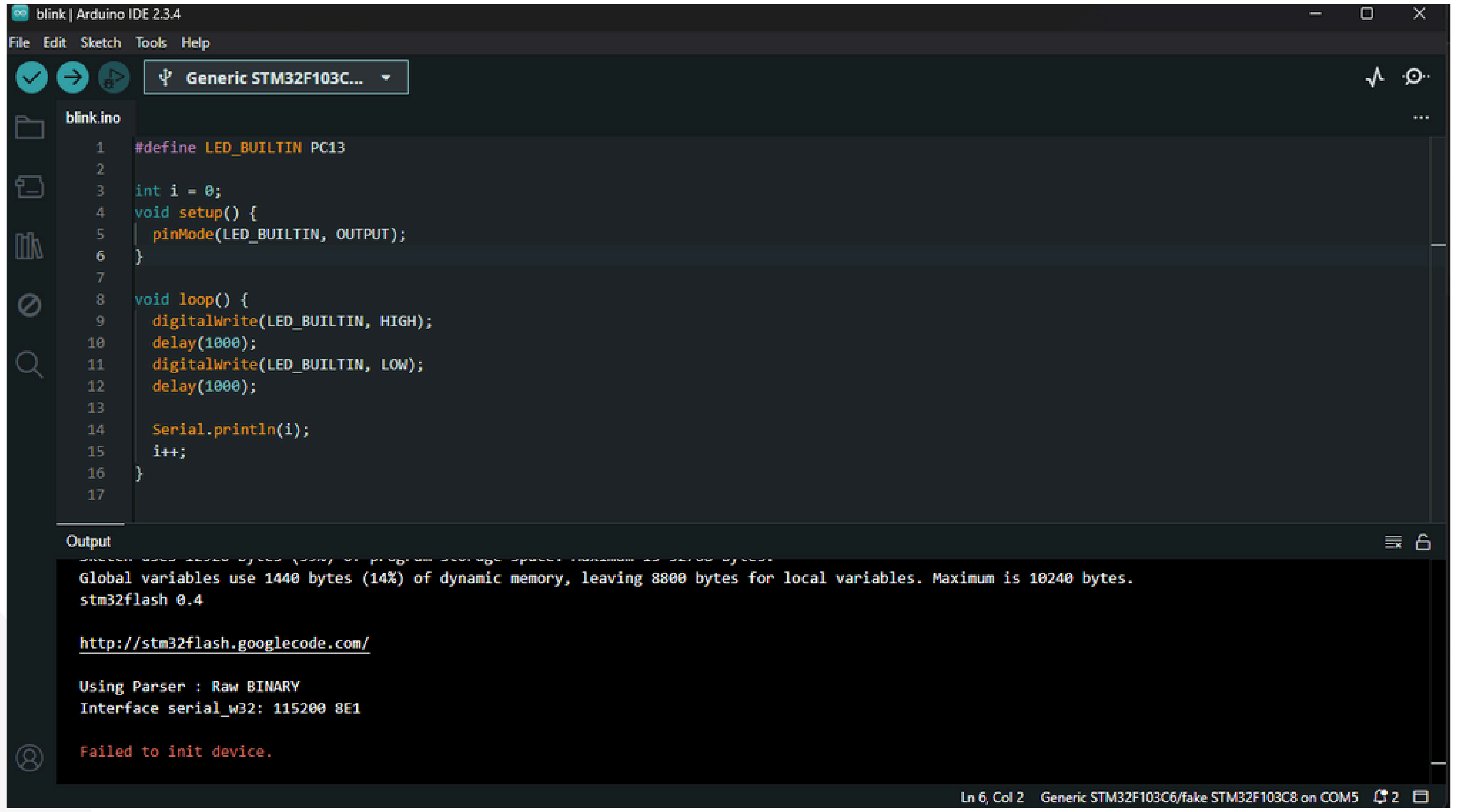

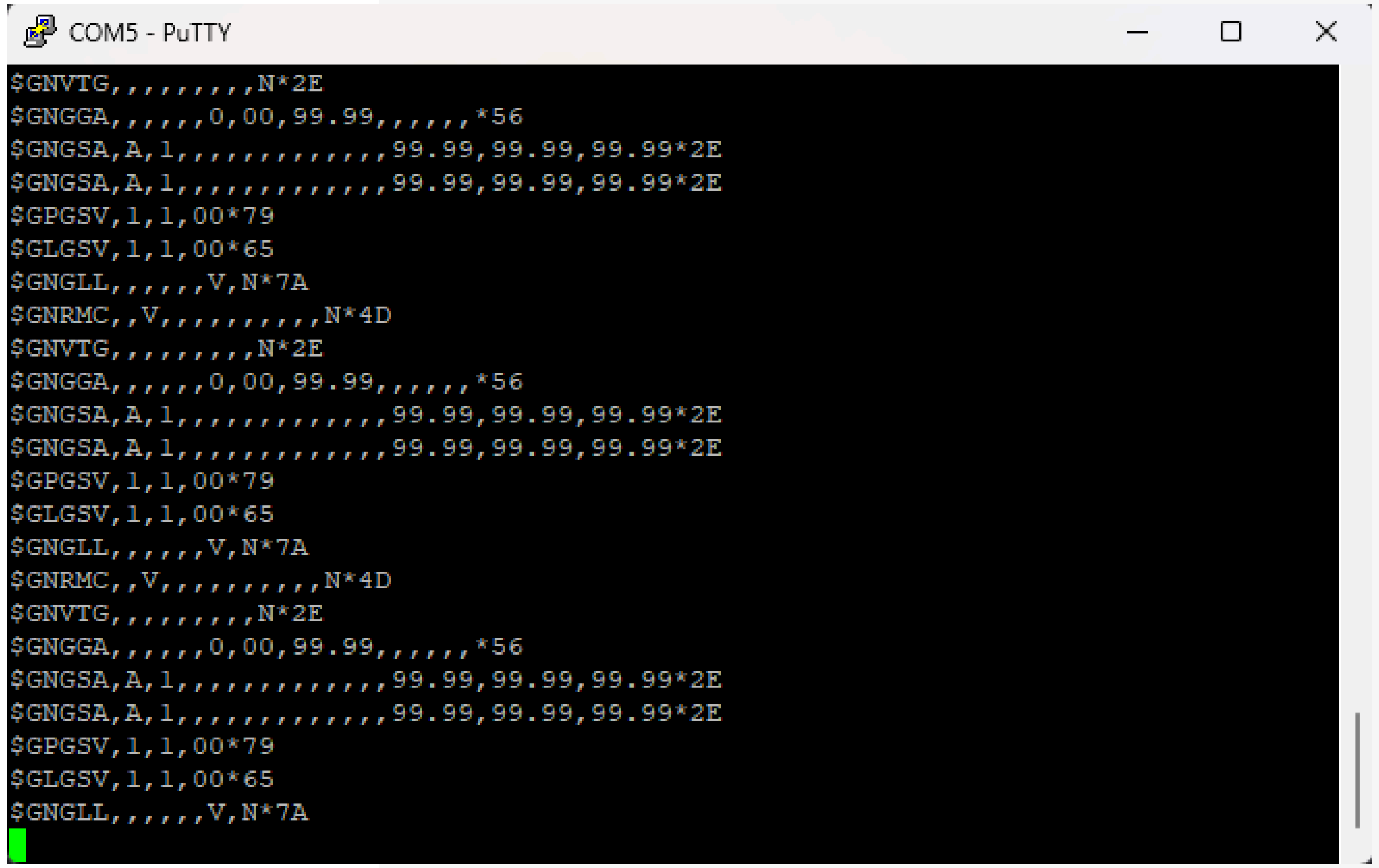

SOCIAL INNOVATION STATEMENT

✓ Problem: High costs & cellular limits exclude many from safety gear

✓ Solution: Low-cost GPS + LoRaWAN—no fees, no cellular needed

✓ Impact: Faster emergency response, fewer preventable injuries

✓ Vision: Scaling safety through resort & community partnerships

CONCLUSION

- ✓ Real-time GPS tracking for skiers of all levels
- ✓ Solves key gaps: Cost, connectivity & usability
- ✓ Future-ready: AI analytics & avalanche detection
- ✓ Proving advanced safety can be accessible

FUTURE IMPROVEMENTS

1. Smarter Detection

- AI motion analysis to reduce false alarms (e.g., falls vs. jumps).
- Avalanche beacon compatibility for backcountry skiers.

2. Better Connectivity

- Mesh networking to improve dead-zone coverage.
- Partner with resorts for LoRaWAN networks in large areas.

3. User Upgrades

- Mobile app for real-time location sharing.
- Vibration alerts for low battery or emergencies.

4. Cost & Sustainability

- Solar-powered charging for longer battery life.
- Reduce costs via bulk manufacturing.

5. Global Expansion

- Adapt for snowmobiling & ice climbing.
- Target high-growth markets (China, Scandinavia).

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THANK YOU

FOR YOUR NICE ATTENTION

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